
Memory-based Self-Explainable Early Exit Networks

Evan Kaiden

University of California Santa Cruz
ekaiden@ucsc.edu

Leilani Gilpin

University of California Santa Cruz
lgilpin@ucsc.edu

Biagio La Rosa

University of California Santa Cruz
bilarosa@ucsc.edu

Introduction Early Exit Neural Networks accelerate inference of deep neural networks by allowing samples to exit early through *internal classifiers* (ICs). Despite recent progress in performance for these methods [1, 2, 3, 4], it remains unexplored whether these early exits can provide interpretability alongside the prediction. Driven by the high demand for trustworthy and interpretable systems, several Self-Explainable (SENN) methods have been proposed for standard classification settings [5, 6, 7]. However, they start from the assumptions that the interpretable layer is attached to the final layer and that the backbone features are representative and well learned, two assumptions that may not hold for exit networks in general. In this work, we propose Memory-based Self-Explainable Early Exit Networks (M-SEEN), a design to make the ICs of exit networks more interpretable. Inspired by [7], M-SEEN uses a sparse content-based attention mechanism to select relevant examples from a memory set to aid and explain the decision process. We show that, compared to alternative SENN designs, M-SEEN best preserves the performance of black-box baselines while providing means to inspect its inner mechanisms.

M-SEEN Let F be a neural network with n layers, x_i the input, f_j the j -th layer, and $F_j(\cdot) = f_j(f_{j-1}(\dots f_1(\cdot)))$ the composition of the first j layers. Standard classification is typically obtained by attaching a classifier f_{cls} to the final hidden representation $F_{n-1}(x_i)$. Exit networks augment this architecture with additional internal classifiers: an IC_j (implemented using either linear, convolutional, or transformer layers [1, 4, 8]) is attached to a hidden layer f_j , producing an early prediction $\hat{y}_i = IC_j(F_j(x_i))$. Taking inspiration from an interpretable classifier designed for small-data settings [7], M-SEEN uses both the latent representation of an input x_i at the layer j and the latent representations of a set of k memory samples, randomly sampled from a set of cached training representations $M = \{m_1, \dots, m_m\}$ at the same layer, to predict $\hat{y}_i$. Specifically, M-SEEN applies sparsemax to the cosine similarity between these representations and uses the resulting scores α as weights to compute a combination of the memory representations: $\mathbf{v}_m = \sum_{m_r \in M_k} \alpha_r m_r$. Finally, a linear classifier predicts $\hat{y}_i = f_{\text{linear}}^j([F_j(x_i), \mathbf{v}_m])$ from the concatenation of the input representation and the weighted memory vector. The idea is that the additional information carried by the memory vector can support the decision process of early layers, while the sparse weights and the selected memory samples at each IC provide insights into the decision process along the network.

Performance To validate our method, we compare its performance against the black-box baseline SDN [1] and two representative SENN designs: LF-CBM [9] and VLG-CBM [10]. SDN was selected because it is a popular baseline for exit networks and its design can be easily extended to self-explainable settings. Conversely, LF-CBM and VLG-CBM represent two of the most recent methods compatible with the exit-network setting, based on the popular framework of concept bottleneck models [6]. We test all competitors on CIFAR-10, CIFAR-100, and TinyImageNet, using WideResNet32-4 as the backbone network at different inference budgets. We follow SDN to place the ICs and to train the backbone and the ICs jointly. Table 1 shows that while LF-CBM and VLG-CBM drop performance by up to 30 percentage points, M-SEEN stays within 1-2% of its black-box

Table 1: Accuracy (%) over 3 different runs reached by competitors across inference budgets. **Bold** indicates the best SENN method while underline indicates the best overall result.

Data	IC-Type	SENN	25%	50%	75%	100%	Max
C10	Black-box	×	<u>87.8</u> ± 0.4	<u>94.3</u> ± 0.1	94.4 ± 0.0	94.4 ± 0.0	94.4 ± 0.1
	M-SEEEN	✓	86.2 ± 0.3	93.9 ± 0.2	94.1 ± 0.2	94.1 ± 0.2	94.1 ± 0.2
	LF-CBM	✓	74.6 ± 0.9	91.8 ± 0.2	94.0 ± 0.1	94.1 ± 0.1	94.1 ± 0.1
	VLG-CBM	✓	72.8 ± 0.4	91.7 ± 0.4	94.4 ± 0.0	94.4 ± 0.0	94.4 ± 0.0
C100	Black-box	×	<u>60.9</u> ± 0.4	<u>74.0</u> ± 0.2	76.1 ± 0.3	75.9 ± 0.2	76.2 ± 0.3
	M-SEEEN	✓	58.7 ± 0.6	72.7 ± 1.1	76.4 ± 0.4	76.2 ± 0.3	76.4 ± 0.4
	LF-CBM	✓	35.5 ± 0.4	57.7 ± 0.7	72.5 ± 0.3	74.7 ± 0.2	74.7 ± 0.2
	VLG-CBM	✓	30.7 ± 0.2	53.8 ± 0.5	72.2 ± 0.1	75.8 ± 0.2	75.8 ± 0.2
T-IM	Black-box	×	<u>40.3</u> ± 0.7	<u>55.7</u> ± 0.5	<u>62.4</u> ± 0.3	<u>62.2</u> ± 0.3	<u>62.7</u> ± 0.3
	M-SEEEN	✓	38.4 ± 0.9	53.4 ± 0.8	61.5 ± 0.3	61.8 ± 0.3	62.4 ± 0.2
	LF-CBM	✓	15.6 ± 1.6	36.2 ± 0.6	53.6 ± 0.5	60.0 ± 0.3	60.0 ± 0.3
	VLG-CBM	✓	18.6 ± 0.9	36.1 ± 0.5	54.6 ± 0.2	61.6 ± 0.2	61.7 ± 0.2

counterpart. We observe that the performance drop of CBM-based methods is large at the earliest exits, where shallow features do not represent abstract concepts yet. Because M-SEEEN compares the input x_i and each memory sample m_r at the same depth, rather than projecting features onto a fixed concept space, selection always operates over the current representation.

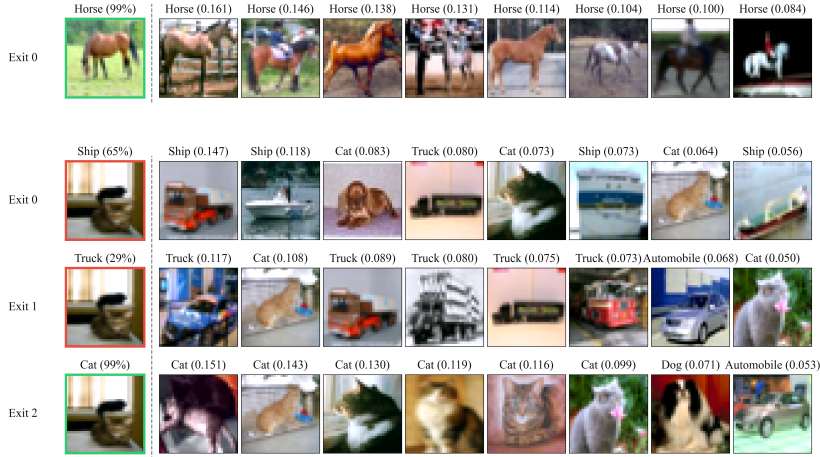


Figure 1: Two sample trajectories. Shown is the predicted class of the input and memory at each exit. The scores correspond to the confidence for the input and the weights for the memory images.

Explanations M-SEEEN helps users analyze the trajectory of samples across the ICs by providing evidence of how and when the model resolves its uncertainty. For example, fig. 1 shows two samples following different trajectories. The first (top) corresponds to an easy sample, as the model exits at the first IC and all the selected memories strongly agree with the input in terms of both visual appearance and class. The second (bottom) represents a more challenging case. Here, the model’s uncertainty is reflected by the conflicting memory samples selected at the first two exits. Note that the relatively high confidence at the first exit may incorrectly suggest that the model is sufficiently confident in its prediction. Conversely, at the third exit, where the representation has become sufficiently rich for the model to predict the correct class, the selected memory samples mostly agree with the predicted class.

Conclusion In this work, we introduced M-SEEEN, a method for aiding and explaining predictions in exit networks. We showed that M-SEEEN performs significantly better than other explainable baselines and showcases an example of how its design can aid interpretability. Future work includes explainable exit strategies and architectural changes to boost performance and explanation quality.

References

- [1] Yigitcan Kaya, Sanghyun Hong, and Tudor Dumitras. Shallow-deep networks: Understanding and mitigating network overthinking. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3301–3310. PMLR, 09–15 Jun 2019.
- [2] Yizeng Han, Dongchen Han, Zeyu Liu, Yulin Wang, Xuran Pan, Yifan Pu, Chao Deng, Junlan Feng, Shiji Song, and Gao Huang. Dynamic perceiver for efficient visual recognition. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5969–5979. IEEE, 2023.
- [3] Gao Huang, Danlu Chen, Tianhong Li, Felix Wu, Laurens van der Maaten, and Kilian Weinberger. Multi-scale dense networks for resource efficient image classification. In *International Conference on Learning Representations*, 2018.
- [4] Maciej Wołczyk, Bartosz Wójcik, Klaudia Bałazy, Igor T Podolak, Jacek Tabor, Marek Śmieja, and Tomasz Trzcinski. Zero time waste: Recycling predictions in early exit neural networks. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 2516–2528. Curran Associates, Inc., 2021.
- [5] Meike Nauta, Jörg Schlötterer, Maurice van Keulen, and Christin Seifert. Pip-net: Patch-based intuitive prototypes for interpretable image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2744–2753, June 2023.
- [6] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5338–5348. PMLR, 13–18 Jul 2020.
- [7] Biagio La Rosa, Roberto Capobianco, and Daniele Nardi. A self-interpretable module for deep image classification on small data. *Applied Intelligence*, aug 2022.
- [8] Arian Bakhtiarnia, Qi Zhang, and Alexandros Iosifidis. Single-layer vision transformers for more accurate early exits with less overhead. *Neural Networks*, 153(C):461–473, September 2022.
- [9] Tuomas Oikarinen, Subhro Das, Lam M. Nguyen, and Tsui-Wei Weng. Label-free concept bottleneck models. In *The Eleventh International Conference on Learning Representations*, 2023.
- [10] Divyansh Srivastava, Ge Yan, and Tsui-Wei Weng. VLG-CBM: Training concept bottleneck models with vision-language guidance. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.